Advance Python Assignment

Topic-7: Inheritance

Q1. Explain different types of inheritance in Python.

Ans.
Inheritance in Python is a concept of object-oriented programming where one class can inherit the properties and methods of another class. It helps in code reuse and makes programs more organized. There are different types of inheritance such as single, multilevel, multiple, hierarchical, and hybrid inheritance. In single inheritance, one class inherits from one parent class. In multilevel inheritance, a class inherits from another class which is already inherited. Multiple inheritance allows a class to inherit from more than one parent class. Hierarchical inheritance means multiple classes inherit from one parent class. Hybrid inheritance is a combination of different types of inheritance.

Example:
```
class A:
   def showA(self):
      print("Class A")

class B(A):
   def showB(self):
      print("Class B")

b1 = B()
b1.showA()
b1.showB()
```

Q2. Explain the use of super() function in inheritance.

Ans.
The super() function in Python is used to access methods and properties of the parent class from the child class. It helps in calling the parent class constructor or methods without directly using the parent class name. This makes the code more flexible and easier to maintain. The super() function is especially useful in inheritance when we want to extend the functionality of the parent class instead of

rewriting the same code.

Example:

```
class Parent:
   def __init__(self):
      print("Parent Constructor")

class Child(Parent):
   def __init__(self):
      super().__init__()
      print("Child Constructor")

c1 = Child()
```